

Remote Controlled Toy Car Demystified

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A remote controlled (RC) car has two small DC motors with tiny gear boxes, one to spin the rear wheels and the other to steer the front wheels. Communication between the car and its hand-held remote control is through a 27MHz radio link. A typical toy car with 27MHz remote control is shown in Fig. 1.

Let us start by looking at the transmitter circuit located inside the remote control, which is shown in Fig. 2. This circuit operates on a 3V battery.



Fig. 1: Typical toy car with 27MHz remote (Credit: amazon.com)

At the heart of the radio transmitter circuit board is 8-pin encoder chip YX4116 (IC1). Rest of the electronics is a 27MHz radio frequency transmitter wired around two transistors and a handful of passive components. There are four control buttons in the 3V (two 1.5V AA battery cells) DC powered transmitter board for left, right, back, and front movement.

Now let's take a look at the schematic of the radio transmitter board QF-1694-5T. Here, the 27MHz carrier

signal is generated by the oscillator comprising XTAL1, transistor T1, and inductor L1. Resistor R2 sets the bias current of T1 to excite the 27MHz (actually 27.145MHz) crystal and gets it oscillating at its natural frequency.

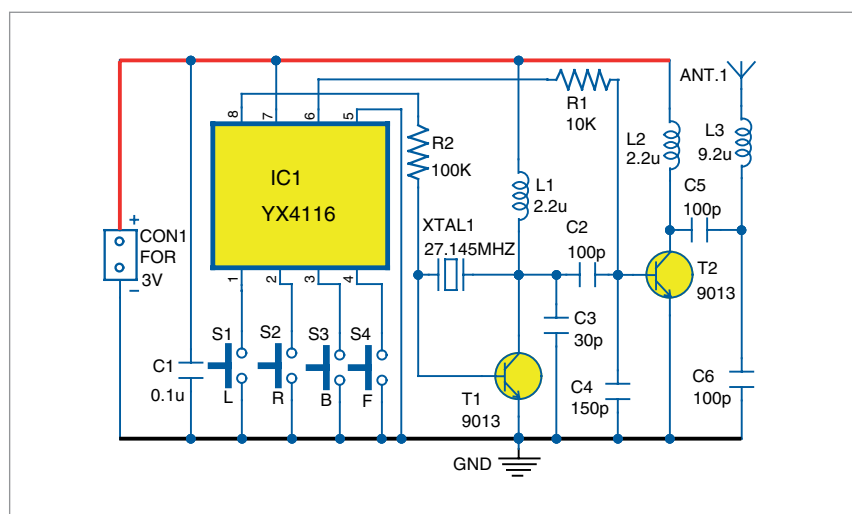


Fig. 2: Circuit diagram of the radio transmitter

PARTS LIST

FOR TRANSMITTER

Semiconductors:

- IC1 - YX4116 radio transmitter
- T1, T2 - 9013 NPN transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 10-kilo-ohm
- R2 - 100-kilo-ohm

Capacitors:

- C1 - 0.1 μ F ceramic disk
- C2, C5, C6 - 100pF ceramic disk
- C3 - 30pF ceramic disk
- C4 - 150pF ceramic disk

Miscellaneous:

- ANT1 - 10cm antenna
- CON1 - 2-pin connector
- S1-S4 - Tactile switch
- L1, L2 - 2.2 μ H inductor
- L3 - 9.2 μ H inductor
- XTAL1 - 27.145MHz crystal
- 3V battery for transmitter

FOR RECEIVER

Semiconductors:

- IC1 - TXM8D423C radio receiver
- T1 - 2SC2712 NPN transistor
- LED1, LED2 - 3mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3-R5, R8 - 2.2-kilo-ohm
- R2, R10 - 220-kilo-ohm
- R6 - 680-ohm
- R7 - 2.2-mega-ohm
- R9 - 56-ohm

Capacitors:

- C1 - 5pF ceramic disk
- C2, C4 - 22pF ceramic disk
- C3, C7, C9, C12, C14 - 100nF ceramic disk
- C5 - 33nF ceramic disk
- C6 - 470pF ceramic disk
- C8, C13 - 2.2nF ceramic disk
- C10 - 4.7 μ F, 16V electrolytic
- C11 - 10 μ F, 16V electrolytic disk

Miscellaneous:

- ANT.1 - 10cm antenna
- CON1 - 2-pin connector
- L1 - K165-103-6T adjustable inductor
- L2 - 2.2 μ H inductor
- M1, M2 - 3V-4.5V DC motor
- S1 - On/off switch
- BATT.1 - 3V-4.5V battery for receiver